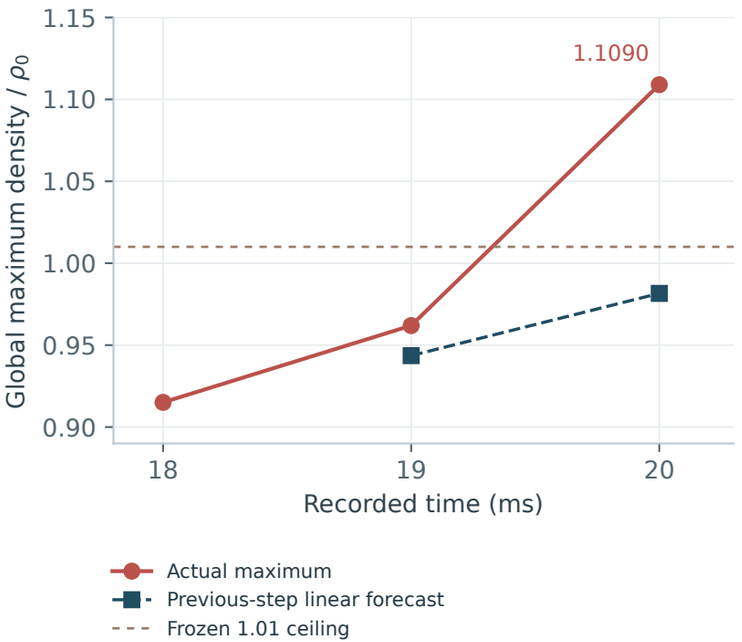
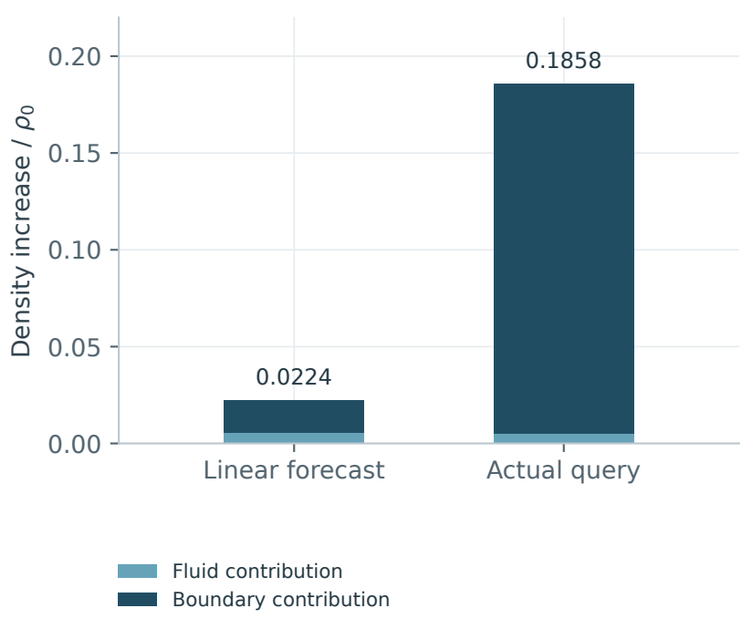
Boundary switching exposes a missed density increase

Exact-distance diagnostic · archived 18–20 ms states · all are unqualified after the first operator-gate failure at 8 ms

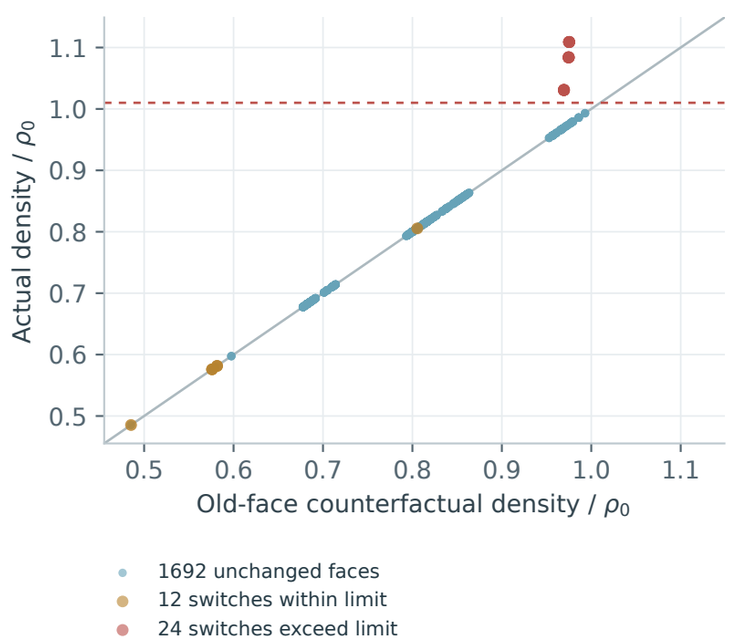
A Forecast versus actual query



B Particle 720, 19 → 20 ms



C Same saved positions, all 1728



A: discrete measurements, not interpolated simulation states. B: the missed increase is dominated by the boundary term.

C: coincident points overlap. Retaining a non-nearest old face underestimates the boundary contribution; this is not a repaired trajectory.

No new solve, integration, particle filtering or threshold change. Exact-volume quadrature alone still leaves all 24 density failures.